

Last class:

- Syllabus/outline
- HW0
- Started Bash:
 - cd, ls, pwd, rm
- editing files with nano

Today:

- More command line
- Redirection
- Shell scripting

At end of last class, created file1.txt and file2.txt and deleted the first:

```
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/writing$ ls
file2.txt  haiku.txt  LittleWomen.txt  thesis
```

Naming things like file2.txt is really bad. Change file names by "moving" the file:

```
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/writing$ ls
file2.txt  haiku.txt  LittleWomen.txt  thesis
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/writing$ mv file2.txt junk_text.txt
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/writing$ ls
haiku.txt  junk_text.txt  LittleWomen.txt  thesis
```

Also use mv to move files to different directories. Use . to indicate current working directory:

```
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/writing/thesis$ mv ../junk_text.txt .
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/writing/thesis$ ls
junk_text.txt
```

Similarly, we can copy files using cp. This creates a duplicate copy, and the 2 files are not linked in any way:

```
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/writing/thesis$ ls
junk_text.txt
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/writing/thesis$ ls ..
haiku.txt  junk_text.txt  LittleWomen.txt  thesis
```

We can copy directories of files recursively (directory and all of its contents) using cp -r:

```
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/writing/thesis$ cp -r ../alkanes .
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/writing/thesis$ ls
alkanes  junk_text.txt
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/writing/thesis$ ls -R
.:
alkanes  junk_text.txt

./alkanes:
cubane.pdb  ethane.pdb  methane.pdb  octane.pdb  pentane.pdb  propane.pdb
```

Cannot use rm on directories directly:

```
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/writing/thesis$ rm alkanes
rm: cannot remove 'alkanes': Is a directory
```

Two ways to do this:

1. If the directory is empty, use rmdir
2. Can always use rm -rf

```
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/writing/thesis$ rmdir alkanes/
rmdir: failed to remove 'alkanes/': Directory not empty
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/writing/thesis$ rm -rf alkanes
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/writing/thesis$ ls
junk_text.txt
```

If you're nervous about accidentally deleting things, use rm -i. If you're extremely careful, you can use rm -f

```
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/writing/thesis$ cp -r ../alkanes .
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/writing/thesis$ rm -rf alkanes
rm: descend into directory 'alkanes'? y
rm: remove regular file 'alkanes/pentane.pdb'? y
rm: remove regular file 'alkanes/cubane.pdb'? y
rm: remove regular file 'alkanes/octane.pdb'? y
rm: remove regular file 'alkanes/ethane.pdb'? y
rm: remove regular file 'alkanes/propane.pdb'? y
rm: remove directory 'alkanes'? y
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/writing/thesis$ cp -r ../alkanes .
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/writing/thesis$ rm -rf alkanes
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/writing/thesis$ ls
junk_text.txt
```

Can copy multiple files into a directory at the same time

```
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/writing/thesis$ cd ..
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/writing$ ls
haiku.txt  junk_text.txt  LittleWomen.txt  thesis
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/writing$ mkdir tmp
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/writing$ cp LittleWomen.txt junk_text.txt tmp
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/writing$ ls -R .
.:
haiku.txt  junk_text.txt  LittleWomen.txt  thesis  tmp

./thesis:
junk_text.txt

./tmp:
junk_text.txt  LittleWomen.txt
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/writing$ rm -rf tmp
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/writing$ cp LittleWomen.txt junk_text.txt file3.txt
cp: target 'file3.txt': No such file or directory
```

When you want to work with many files at once, use * and ? as wildcards:

- ? represents exactly one character
- * represents zero or more characters

```
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/alkanes$ ls
cubane.pdb  ethane.pdb  methane.pdb  octane.pdb  pentane.pdb  propane.pdb
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/alkanes$ ls *.pdb
cubane.pdb  ethane.pdb  methane.pdb  octane.pdb  pentane.pdb  propane.pdb
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/alkanes$ ls ?th*
ethane.pdb
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/alkanes$ ls ?*th*
methane.pdb
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/alkanes$ ls ?c*
ls: cannot access '?c*': No such file or directory
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/alkanes$ ls ?c*
octane.pdb
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/alkanes$ ls *c*
cubane.pdb  octane.pdb
```

The really cool stuff happens when we start to combine commands

Use wc as an example (word count):

- prints lines, words, characters in a file or files

```
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/alkanes$ wc cubane.pdb
20  156 1158 cubane.pdb
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/alkanes$ wc *.pdb
20  156 1158 ethane.pdb
12   84  622 ethane.pdb
9   57  422 methane.pdb
30  246 1828 octane.pdb
21  165 1226 pentane.pdb
15  111  825 propane.pdb
107  819 6081 total
```

Can use -l or -w or -m to get just one of these

```
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/alkanes$ wc -w *.pdb
156 cubane.pdb
84 ethane.pdb
57 methane.pdb
246 octane.pdb
165 pentane.pdb
111 propane.pdb
819 total
```

Note that wc needs a file name:

```
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/alkanes$ wc -l
^C
```

Use ctrl-c to cancel

It was waiting for input -l could just type, use ctrl-d to end input

```
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/alkanes$ wc -l
sdhjadfjhkfasjkh
afhjafkuyafeku
afhfdsgjdsd
acruuuiaew
4
```

Sometimes we want to capture output in a file. Bash provides a nice way to do this with redirection

```
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/alkanes$ wc -l *.pdb
20 cubane.pdb
12 ethane.pdb
9 methane.pdb
30 octane.pdb
21 pentane.pdb
15 propane.pdb
107 total
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/alkanes$ wc -l b* > lengths.txt
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/alkanes$ ls -ltr
total 28
-rw-r--r-- 1 scottmac scottmac 825 Mar 12 2023 propane.pdb
-rw-r--r-- 1 scottmac scottmac 1226 Mar 12 2023 pentane.pdb
-rw-r--r-- 1 scottmac scottmac 1828 Mar 12 2023 octane.pdb
-rw-r--r-- 1 scottmac scottmac 422 Mar 12 2023 methane.pdb
-rw-r--r-- 1 scottmac scottmac 622 Mar 12 2023 ethane.pdb
-rw-r--r-- 1 scottmac scottmac 1158 Mar 12 2023 cubane.pdb
-rwxr-xr-x 1 scottmac scottmac 110 Sep 11 10:50 get_headers.txt
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/alkanes$ cat lengths.txt
20 cubane.pdb
12 ethane.pdb
9 methane.pdb
30 octane.pdb
21 pentane.pdb
15 propane.pdb
107 total
```

Note that cat (short for concatenate) to display files to the screen

Four variants on redirection:

1. command > file.txt redirects output from command to file.txt, overwriting any content already there
2. command >> file.txt redirects output and appends to end of file.txt if it already exists.
3. command &> file.txt redirects both error messages and output, overwriting file.txt if it exists.
4. command &>> file.txt redirects both error messages and output, appending to file.txt if it exists.

```
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/alkanes$ wc -l b* > lengths.txt
wc: 'b*': No such file or directory
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/alkanes$ wc -l p* > lengths.txt
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/alkanes$ cat lengths.txt
21 pentane.pdb
15 propane.pdb
36 total
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/alkanes$ wc -l b* &> lengths.txt
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/alkanes$ cat lengths.txt
wc: 'b*': No such file or directory
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/alkanes$ wc -l p* >> lengths.txt
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/alkanes$ cat lengths.txt
wc: 'b*': No such file or directory
21 pentane.pdb
15 propane.pdb
36 total
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/alkanes$ wc -l b* &>> lengths.txt
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/alkanes$ cat lengths.txt
wc: 'b*': No such file or directory
21 pentane.pdb
15 propane.pdb
36 total
wc: 'b*': No such file or directory
```

Redirection is very useful when you run software on remote machines or supercomputers. Typically we send jobs to a queue where they are run when resources are available. In that setting, cannot simply watch output as it doesn't run interactively.

With large data files, sometimes need to sort our output. sort command does that

```
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data$ cat numbers.txt
10
2
19
22
6
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data$ sort numbers.txt
10
19
22
6
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data$ sort -n numbers.txt
2
6
10
19
22
```

By default, sort is alphabetic, so 10 comes before 5. For numerical sort, use sort -n. Notice:

```
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/alkanes$ sort -bn lengths.txt
9 methane.pdb
12 ethane.pdb
15 propane.pdb
20 cubane.pdb
21 pentane.pdb
30 octane.pdb
107 total
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/alkanes$ wc -l * | sort -n
7 lengths.txt
9 methane.pdb
12 ethane.pdb
15 propane.pdb
20 cubane.pdb
21 pentane.pdb
30 octane.pdb
114 total
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/alkanes$ wc -l * | sort -n | tail -n 1
114 total
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/alkanes$ wc -l * | sort -n | head -n 2
7 lengths.txt
9 methane.pdb
```

head shows us the first lines of a file (or whatever its input is). By default, shows first 10 lines, use -n to specify how many lines to show. tail works the same way, but shows last lines.

There are many powerful tools in bash. Nowadays its much more common to code complex tasks on files using python (or similar languages), but writing "shell scripts" is still fully supported

IMPORTANT: Anything you do that is a critical step in your research should be stored in a file, as should its inputs and outputs. File storage is cheap, your time to recreate data is not!

Example: Suppose you want to grab data from header of each file:

```
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/alkanes$ cat ethane.pdb
COMPND      ETHANE
AUTHOR      DAVE WOODCOCK   95 12 18
ATOM        1 C             1      -0.752   0.001  -0.141  1.00  0.00
ATOM        2 C             1      0.752  -0.001   0.141  1.00  0.00
ATOM        3 H             1     -1.158   0.991   0.000  1.00  0.00
ATOM        4 H             1     -1.240  -0.737   0.496  1.00  0.00
ATOM        5 H             1     -0.924  -0.249  -1.188  1.00  0.00
ATOM        6 H             1     1.158  -0.991  -0.070  1.00  0.00
ATOM        7 H             1     0.924   0.249   1.188  1.00  0.00
ATOM        8 H             1     1.240   0.737  -0.496  1.00  0.00
TER         9              1
END
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/alkanes$ for file in *.pdb
> do
>   head -n 2 $file | cat
> done
COMPND      CUBANE
AUTHOR      DAVE WOODCOCK   95 12 06
COMPND      ETHANE
AUTHOR      DAVE WOODCOCK   95 12 18
COMPND      METHANE
AUTHOR      DAVE WOODCOCK   95 12 18
COMPND      OCTANE
AUTHOR      DAVE WOODCOCK   96 01 05
COMPND      PENTANE
AUTHOR      DAVE WOODCOCK   95 12 18
COMPND      PROPANE
AUTHOR      DAVE WOODCOCK   95 12 18
```

We just wrote a for loop at the command line. This works, but

1. Doesn't save data in a file
2. Have to reconstruct command if you want to rerun it.

Better to save command to a file:

```
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/alkanes$ nano get_headers.sh
rm -rf headers.txt
for file in *.pdb
do
  head -n 2 $file >> headers.txt
done
```

```
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/alkanes$ bash get_headers.sh
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/alkanes$ chmod u+x get_headers.sh
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/alkanes$ ./get_headers.sh
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/alkanes$ ls -ltr
total 36
-rw-r--r-- 1 scottmac scottmac 825 Mar 12 2023 propane.pdb
-rw-r--r-- 1 scottmac scottmac 1226 Mar 12 2023 pentane.pdb
-rw-r--r-- 1 scottmac scottmac 1828 Mar 12 2023 octane.pdb
-rw-r--r-- 1 scottmac scottmac 422 Mar 12 2023 methane.pdb
-rw-r--r-- 1 scottmac scottmac 622 Mar 12 2023 ethane.pdb
-rw-r--r-- 1 scottmac scottmac 1158 Mar 12 2023 cubane.pdb
-rwxr-xr-x 1 scottmac scottmac 110 Sep 11 11:29 get_headers.sh
-rw-rw-r-- 1 scottmac scottmac 334 Sep 11 11:29 headers.txt
```

Notes:

1. Use which to find out which version of a function/shell script/command you're using
2. Add a shell to shell script using #! <shell> <u>x</u>
3. chmod to "change file mode". Syntax is a bit complex

Can also write scripts that take command line options:

```
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/alkanes$ which bash
/usr/bin/bash
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/alkanes$ nano get_headers.sh
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/alkanes$ chmod u+x get_headers.sh
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/alkanes$ ./get_headers.sh
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/alkanes$ ls -ltr
total 36
-rw-r--r-- 1 scottmac scottmac 825 Mar 12 2023 propane.pdb
-rw-r--r-- 1 scottmac scottmac 1226 Mar 12 2023 pentane.pdb
-rw-r--r-- 1 scottmac scottmac 1828 Mar 12 2023 octane.pdb
-rw-r--r-- 1 scottmac scottmac 422 Mar 12 2023 methane.pdb
-rw-r--r-- 1 scottmac scottmac 622 Mar 12 2023 ethane.pdb
-rw-r--r-- 1 scottmac scottmac 1158 Mar 12 2023 cubane.pdb
-rwxr-xr-x 1 scottmac scottmac 110 Sep 11 11:29 get_headers.sh
-rw-rw-r-- 1 scottmac scottmac 334 Sep 11 11:29 headers.txt
```

1. Use which to find out which version of a function/shell script/command you're using
2. Add a shell to shell script using #! <shell> <u>x</u>
3. chmod to "change file mode". Syntax is a bit complex

Can also write scripts that take command line options:

```
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/alkanes$ cat get_headers.sh
#!/usr/bin/bash
rm -rf headers.txt
for file in $@
do
  head -n 2 $file >> headers.txt
done

scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/alkanes$ ./get_headers.sh *.pdb
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/alkanes$ cat headers.txt
COMPND      CUBANE
AUTHOR      DAVE WOODCOCK   95 12 06
COMPND      ETHANE
AUTHOR      DAVE WOODCOCK   95 12 18
COMPND      METHANE
AUTHOR      DAVE WOODCOCK   95 12 18
COMPND      OCTANE
AUTHOR      DAVE WOODCOCK   96 01 05
COMPND      PENTANE
AUTHOR      DAVE WOODCOCK   95 12 18
COMPND      PROPANE
AUTHOR      DAVE WOODCOCK   95 12 18
```

Use \$@ for all command line arguments, \$1 for first, \$2 for second, etc. Be careful here, counting is after all wildcard arguments are expanded.

Navigating file systems is an important task. We've already seen ls -R to recursively list files. A similar command is find

```
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/alkanes$ cd ..
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data$ ls 0R
ls: cannot access '0R': No such file or directory
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data$ ls -R
.:
alkanes  animal-counts  creatures  numbers.txt  project  writing

./alkanes:
cubane.pdb  get_headers.sh  lengths.txt  octane.pdb  propane.pdb
ethane.pdb  headers.txt    methane.pdb  pentane.pdb

./animal-counts:
animals.csv

./creatures:
basilisk.dat  minotaur.dat  unicorn.dat

./project:
data  results

./project/data:

./project/results:

./writing:
haiku.txt  junk_text.txt  LittleWomen.txt  thesis

./writing/thesis:
junk_text.txt
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data$ find .
.
./alkanes
./alkanes/pentane.pdb
./alkanes/get_headers.sh
./alkanes/lengths.txt
./alkanes/cubane.pdb
./alkanes/octane.pdb
./alkanes/ethane.pdb
./alkanes/propane.pdb
./alkanes/methane.pdb
./alkanes/headers.txt
./writing
./writing/LittleWomen.txt
./writing/haiku.txt
./writing/junk_text.txt
./writing/thesis/junk_text.txt
./numbers.txt
./project
./project/data
./project/results
./creatures/basilisk.dat
./creatures/minotaur.dat
./creatures/unicorn.dat
./animal-counts
./animal-counts/animals.csv
```

These show us what files are there, but not what's in them. For searching within the files, use grep

```
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data$ cd writing/
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/writing$ ls
haiku.txt  junk_text.txt  LittleWomen.txt  thesis
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/writing$ grep elephant LittleWomen.txt
an elephant in silk and lace.
    With elephantine grace,
    young gentlemen would have liked better, but elephants could not have
```

Output shows us lines that match given word. Sometimes useful to be able to get line numbers:

```
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/writing$ wc -l LittleWomen.txt
21022 LittleWomen.txt
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/writing$ grep -n elephant LittleWomen.txt
3022: an elephant in silk and lace.
4322:  With elephantine grace,
17179: young gentlemen would have liked better, but elephants could not have
```

Note that this includes "elephantine" and "elephants", because they both contain the string "elephant". To exclude these, use -w (for word boundary)

```
scottmac@kaczmarz:~/CMS6950/shell-lesson-data/exercise-data/writing$ grep -nw elephant LittleWomen.txt
3822: an elephant in silk and lace.
```